Submission2

Zhiying(Winnie)Chen

2026-11-02

My Choice of Genes

- Genes: GNLY, LYN, IL26
- Continuous covariate: mat_vitD(vitamin D, ng/mL)
- Categorical covariate 1: bmi_group
- Cateforical covariate 2: preg_preeclampsia (Normal vs. Preeclampsia)

Why these tow new genes? I screened the genes that are in the expression file for an association with vitamin D and with preeclampsia, and picked two that behave differently from GNLY. * LYN goes up with vitamin D (GNLY goes down) and is expressed at a high level. * IL26 is expressed at a very low level, so its y-axis is on a completely different scale.

Step1. Load the raw data

```
#read.csv with row.names = 1 uses the first column (probe IDs) as row names
covar <- read.csv(file = 'GSE85307_covar-1.csv')
geneNames <- read.csv(file = 'GSE85307_geneNames.csv')
expr <- read.csv(file = 'GSE85307_geneExpression.csv', row.names = 1)

# The first column name of the covariate file can come in with a hidden extra
# character (BOM), so we rename it ourselves
colnames(covar)[1] <- 'subjectID'

# Look at the the structure
str(covar)
```

```
## 'data.frame':    157 obs. of  10 variables:
##  $ subjectID                  : chr  "6053887-M" "3064425-M" "6094898-M" "7050617-M" ...
##  $ geoID                      : chr  "GSM2264346" "GSM2264347" "GSM2264348" "GSM2264349" ...
##  $ mat_age_yrs                : chr  "age_years: 31.06639288" "age_years: 27.79466119" "age_years: 3
##  $ preg_preeclampsia          : chr  "pregnancy_condition: Preeclampsia" "pregnancy_condition: Norma
##  $ delivery_gestAge_wks       : chr  "gestation_weeks: 37.42857143" "gestation_weeks: 39.42857143"
##  $ sampleCollection_gestAge_wks: chr  "blood_draw_at_gestation_week: 10" "blood_draw_at_gestation_we
##  $ batch_Microarray           : chr  "microarray_hybridization_date: 03/15/2013" "microarray_hybridi
##  $ mat_BMI                    : chr  "bmi_kg/m2: 37.52407538" "bmi_kg/m2: 20.59144273" "bmi_kg/m2: 1
##  $ mat_vitD                   : chr  "vitamind_25(0h)_ng/ml_whole_blood_baseline: 7.94" "vitamind_25
##  $ mat_asthma                 : chr  "asthma_0=healthy_1=asthmatic: 1" "asthma_0=healthy_1=asthmatic
```

```r
dim(expr)
```

```
## [1] 500 157
```

```r
head(geneNames)
```

```
##   PROBEID      GENE
## 1 8128043      CNR1
## 2 7962489 PLEKHA8P1
## 3 7920271    S100A4
## 4 7900468      NFYC
## 5 8026496 LINC00661
## 6 7989725     PDCD7
```

Step 2. Clean the covariates

Every covariate value is stored as text with a label, like "age_year: xxx". I remove everything up to and including ":" and then fix the data types.

```r
# Loop through columns 3 to 10 (all covariates; columns 1-2 are the IDs)
# '.*: ' means "any characters followed by a colon and a space"
for(i in colnames(covar)[3:10]){
  covar[,i] <- sub(pattern = '.*: ',replacement = '', x = covar[,i])
}
head(covar)
```

```
##   subjectID      geoID mat_age_yrs preg_preeclampsia delivery_gestAge_wks
## 1 6053887-M GSM2264346 31.06639288      Preeclampsia          37.42857143
## 2 3064425-M GSM2264347 27.79466119            Normal          39.42857143
## 3 6094898-M GSM2264348 34.31895962            Normal          38.85714286
## 4 7050617-M GSM2264349 33.42642026            Normal          39.14285714
## 5 7016015-M GSM2264350 28.27926078      Preeclampsia          40.14285714
## 6 7054360-M GSM2264351 23.50171116            Normal          41.14285714
##   sampleCollection_gestAge_wks batch_Microarray     mat_BMI mat_vitD mat_asthma
## 1                           10       03/15/2013 37.52407538     7.94          1
## 2                  17.42857143       03/15/2013 20.59144273     40.3          1
## 3                           13       03/15/2013 20.71118512     40.8          1
## 4                  17.28571429       03/15/2013 31.99358316     52.2          0
## 5                  10.71428571       03/15/2013 27.17734484     22.8          0
## 6                  10.57142857       03/19/2013 22.28684666     44.6          0
```

```r
# Numeric variables
covar$mat_age_yrs <- as.numeric(covar$mat_age_yrs)
covar$delivery_gestAge_wks <- as.numeric(covar$delivery_gestAge_wks)
covar$sampleCollection_gestAge_wks <- as.numeric(covar$sampleCollection_gestAge_wks)
covar$mat_BMI <- as.numeric(covar$mat_BMI)
covar$mat_vitD <- as.numeric(covar$mat_vitD)

# Categorical variables
covar$preg_preeclampsia <- factor(covar$preg_preeclampsia,
                                  levels = c('Normal','Preeclampsia'))
```

```
covar$mat_asthma <- factor(ifelse(covar$mat_asthma=='1','Asthma','Healthy'))

# Check that the types and values look right
str(covar)
```

```
## 'data.frame':    157 obs. of  10 variables:
##  $ subjectID                   : chr  "6053887-M" "3064425-M" "6094898-M" "7050617-M" ...
##  $ geoID                       : chr  "GSM2264346" "GSM2264347" "GSM2264348" "GSM2264349" ...
##  $ mat_age_yrs                 : num  31.1 27.8 34.3 33.4 28.3 ...
##  $ preg_preeclampsia           : Factor w/ 2 levels "Normal","Preeclampsia": 2 1 1 1 2 1 1 1 1 1 ...
##  $ delivery_gestAge_wks        : num  37.4 39.4 38.9 39.1 40.1 ...
##  $ sampleCollection_gestAge_wks: num  10 17.4 13 17.3 10.7 ...
##  $ batch_Microarray            : chr  "03/15/2013" "03/15/2013" "03/15/2013" "03/15/2013" ...
##  $ mat_BMI                     : num  37.5 20.6 20.7 32 27.2 ...
##  $ mat_vitD                    : num  7.94 40.3 40.8 52.2 22.8 44.6 26.2 21.8 39.2 42.9 ...
##  $ mat_asthma                  : Factor w/ 2 levels "Asthma","Healthy": 1 1 1 2 2 2 1 1 2 1 ...
```

```
summary(covar)
```

```
##      subjectID         geoID      mat_age_yrs       preg_preeclampsia
##  Length   :157   Length   :157   Min.   :18.43   Normal      :110
##  N.unique :157   N.unique :157   1st Qu.:22.15   Preeclampsia: 47
##  N.blank  :  0   N.blank  :  0   Median :26.57
##  Min.nchar:  9   Min.nchar: 10   Mean   :26.38
##  Max.nchar:  9   Max.nchar: 10   3rd Qu.:30.14
##                                  Max.   :38.76
##  delivery_gestAge_wks sampleCollection_gestAge_wks  batch_Microarray
##  Min.   :26.43        Min.   :10.00                 Length   :157
##  1st Qu.:38.14        1st Qu.:12.14                 N.unique :  8
##  Median :39.29        Median :13.43                 N.blank  :  0
##  Mean   :38.82        Mean   :14.13                 Min.nchar: 10
##  3rd Qu.:40.00        3rd Qu.:16.43                 Max.nchar: 10
##  Max.   :42.00        Max.   :18.86
##     mat_BMI         mat_vitD       mat_asthma
##  Min.   :17.95   Min.   : 5.11   Asthma :63
##  1st Qu.:22.90   1st Qu.:13.40   Healthy:94
##  Median :27.33   Median :20.40
##  Mean   :28.94   Mean   :22.98
##  3rd Qu.:33.27   3rd Qu.:29.50
##  Max.   :61.95   Max.   :80.80
```

Step 3. Derive a categorical variable from a continuous one

```
# BMI groups: <25, 25-30, >=30
# Only 5 women have the BMI < 18.5, too few for their own group, so they are
# combined with the "<25" group
covar$bmi_group <- cut(covar$mat_BMI,
                       breaks = c(0,25,30,Inf),
                       labels = c('<25','25-30','>=30'))

# Check the sample sizes
table(covar$bmi_group,covar$preg_preeclampsia)
```

```
##
##         Normal Preeclampsia
##   <25       40           13
##   25-30     39            8
##   >=30      31           26
```

Step 4. Build one data frame with all three genes

```
genesOfInterest <- c('GNLY','LYN','IL26')

# Gene name <- probe ID
probeLookup <- geneNames[geneNames$GENE %in% genesOfInterest,]
probeLookup$PROBEID <- as.character(probeLookup$PROBEID)

# Make sure there is exactly one probe per gene and all are in the
# expression data (all True). Only 500 of the 750 genes in geneNames are in expr.
nrow(probeLookup)
```

```
## [1] 3
```

```
table(probeLookup$PROBEID %in% rownames(expr))
```

```
##
## TRUE
##    3
```

```
# Transpose so smaples are rows
exprT <- t(expr)

# Startt from the covariates and add one column per gene
plotData <- covar
for(i in genesOfInterest){
  # Find the probe ID for this gene
  probeID <- probeLookup[probeLookup$GENE==i,'PROBEID']
  # Pull out the gene's values, ordering the samples by the covariate geoIDs
  # so that each row of the covariates gets its own sample's expression
  plotData[,i] <- as.numeric(exprT[plotData$geoID,probeID])
}

# Check there are no missing values (all F)
table(is.na(plotData[,genesOfInterest]))
```

```
##
## FALSE
##   471
```

```
# Spot check one sample for each gene against the raw expression table(all T)
for(i in genesOfInterest){
  probeID <- probeLookup[probeLookup$GENE==i,'PROBEID']
  print(expr[probeID,plotData$geoID[1]]==plotData[1,i])
}
```

```
## [1] TRUE
## [1] TRUE
## [1] TRUE
```

```
head(plotData[,c('geoID','mat_vitD','bmi_group','preg_preeclampsia',genesOfInterest)])
```

```
##        geoID mat_vitD bmi_group preg_preeclampsia      GNLY      LYN     IL26
## 1 GSM2264346     7.94      >=30      Preeclampsia  9.575427 12.28095 4.148253
## 2 GSM2264347    40.30       <25            Normal  9.572921 12.40005 3.976949
## 3 GSM2264348    40.80       <25            Normal 10.165941 12.28026 3.967448
## 4 GSM2264349    52.20      >=30            Normal  9.453005 12.13382 3.877176
## 5 GSM2264350    22.80     25-30      Preeclampsia 10.406808 12.20479 3.898191
## 6 GSM2264351    44.60       <25            Normal  9.330544 12.19434 3.924671
```

Step 5. The function

```
plotGeneExpression <- function(data,geneList,contVar,catVars) {

  # Define color palette
  colorPalette <- c('grey50','darkgreen')

  # Define theme
  myTheme <- theme_classic(base_size = 14) +
    theme(legend.position = 'top',
          plot.title = element_text(size = 14),
          plot.title.position = 'plot')

  # Define somewhere to store the plots
  myPlots <- list()

  # Loop through all genes
  for (i in geneList) {

    # Plot 1: histogram of gene expression
    # (.data[[i]] turns the gene name, which is text, into the column of that name)
    p1 <- ggplot(data,aes(x = .data[[i]])) +
      geom_histogram(bins = 20,fill = 'darkgreen',color = 'white') +
      labs(title = paste('Distribution of',i,'Expression'),
           x = paste(i,'Expression'),y = 'Number of Samples') +
      myTheme

    # Plot 2: continuous covariate vs. gene expression
    p2 <- ggplot(data,aes(x = .data[[contVar]],y = .data[[i]])) +
      geom_point() +
      # Add linear fit with shaded error
      geom_smooth(method = lm,formula = y ~ x,color = 'darkgreen',fill = 'darkgreen') +
      # Leave empty space at the top of the plot for the correlation
      scale_y_continuous(expand = expansion(mult = c(0.05,0.2))) +
      # Add the Pearson correlation
      ggpubr::stat_cor(method = 'pearson') +
      labs(title = paste(i,'Expression vs.',contVar),
```

```
          x = contVar,y = paste(i,'Expression')) +
      myTheme

    # Plot 3: boxplot by both categorical variables
    p3 <- ggplot(data,aes(x = .data[[catVars[1]]],y = .data[[i]],
                          fill = .data[[catVars[2]]])) +
      # outlier.shape = NA because every sample is drawn as a point below
      geom_boxplot(outlier.shape = NA) +
      geom_point(position = position_jitterdodge(jitter.width = 0.15)) +
      scale_fill_manual(values = colorPalette) +
      labs(title = str_wrap(paste(i,'Expression by',catVars[1],'and',catVars[2]),
                            width = 45),
           x = catVars[1],y = paste(i,'Expression'),fill = catVars[2]) +
      myTheme

    # Store the 3 plots for this gene
    myPlots[[i]] <- list('histogram' = p1,'scatter' = p2,'boxplot' = p3)
  }

  # Return the list of plots
  myPlots
}
```

Step 6. Run the function

```
# One call makes all 9 plots (3 plots x 3 genes)
genePlots <- plotGeneExpression(data = plotData,
                                geneList = genesOfInterest,
                                contVar = 'mat_vitD',
                                catVars = c('bmi_group','preg_preeclampsia'))

# Look at the structure of what the function returned
names(genePlots)
```

```
## [1] "GNLY" "LYN"  "IL26"
```

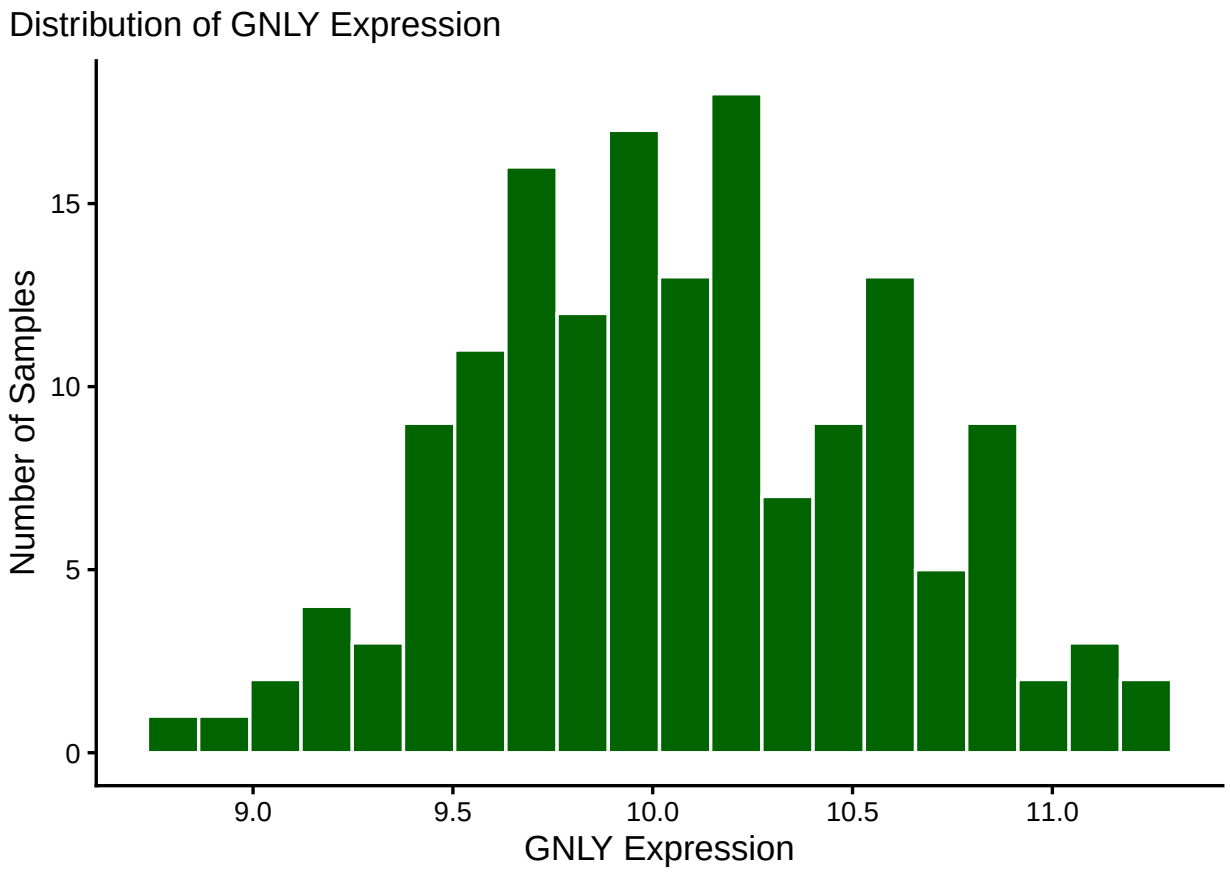
```
names(genePlots$GNLY)
```

```
## [1] "histogram" "scatter"   "boxplot"
```

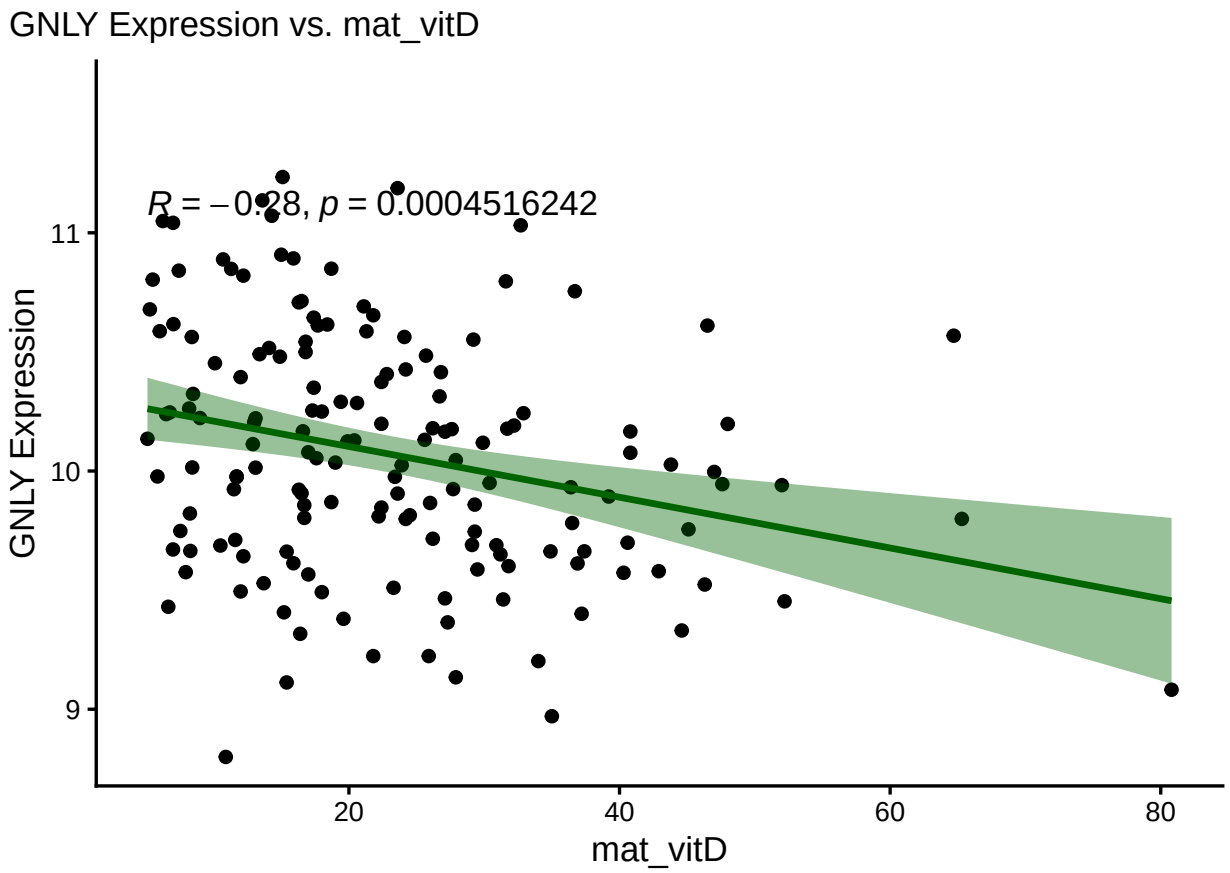
```
# Save every plot, e.g. submission2_GNLY_histogram.png
for (i in names(genePlots)) {
  for (j in names(genePlots[[i]])) {
    ggsave(filename = paste0('submission2_',i,'_',j,'.png'),
           plot = genePlots[[i]][[j]],width = 7,height = 5)
  }
}
```

Plots for GNLY

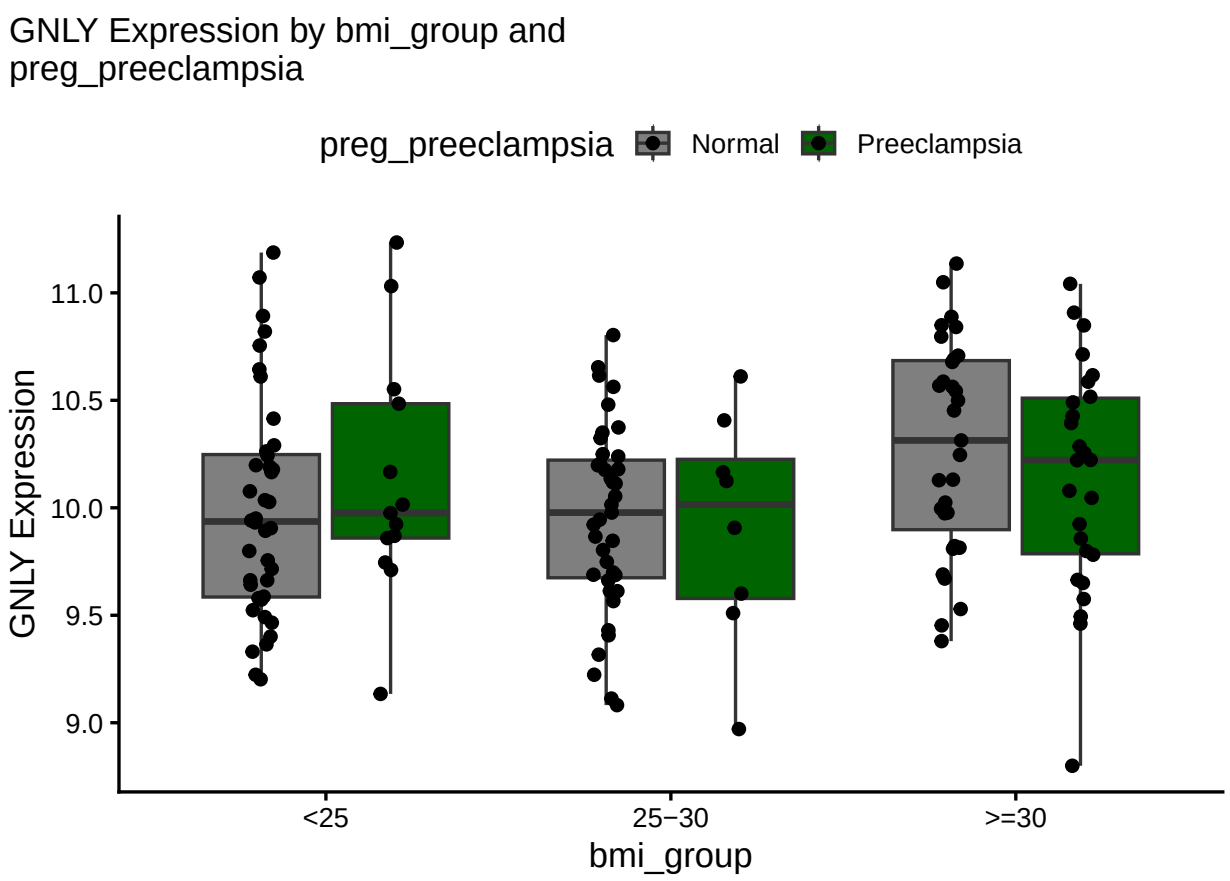
```
genePlots$GNLY$histogram
```




```
genePlots$GNLY$scatter
```

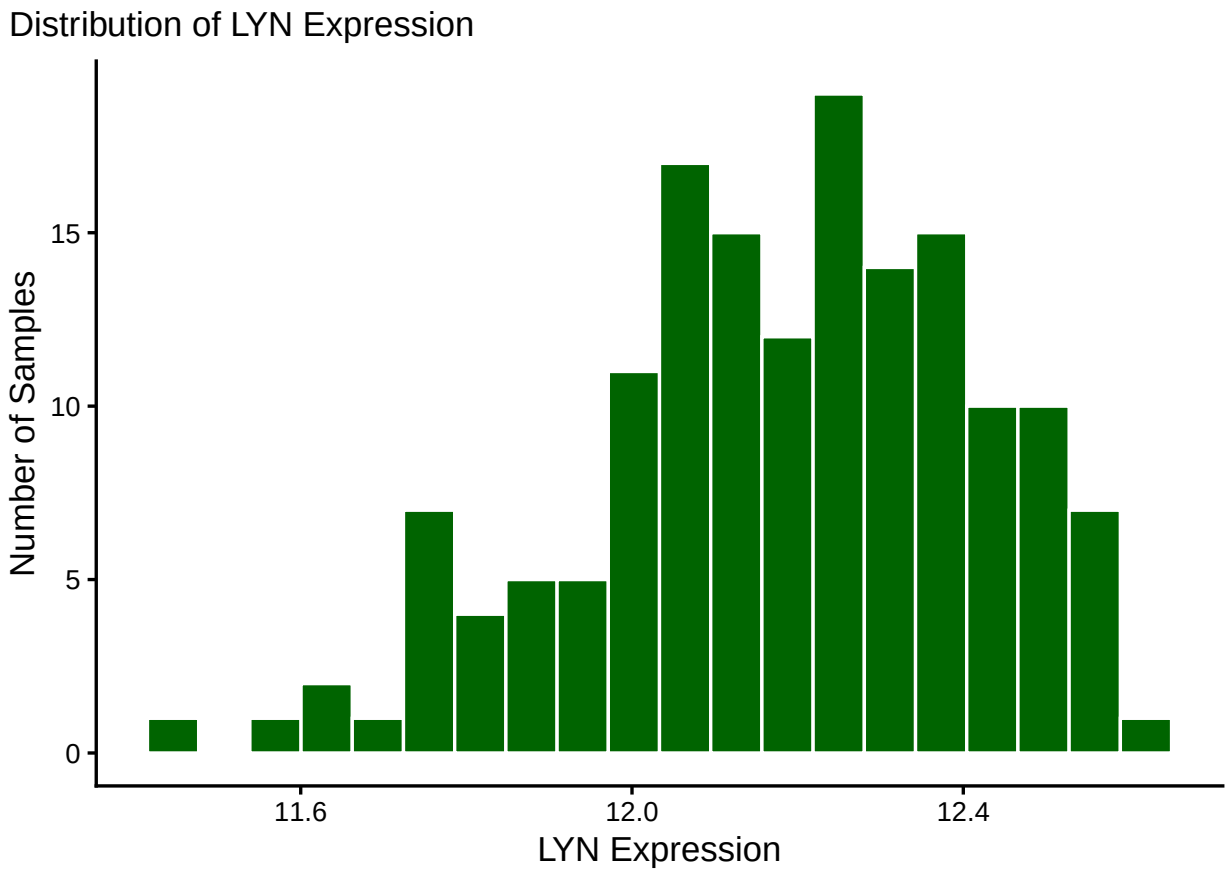



```
genePlots$GNLY$boxplot
```

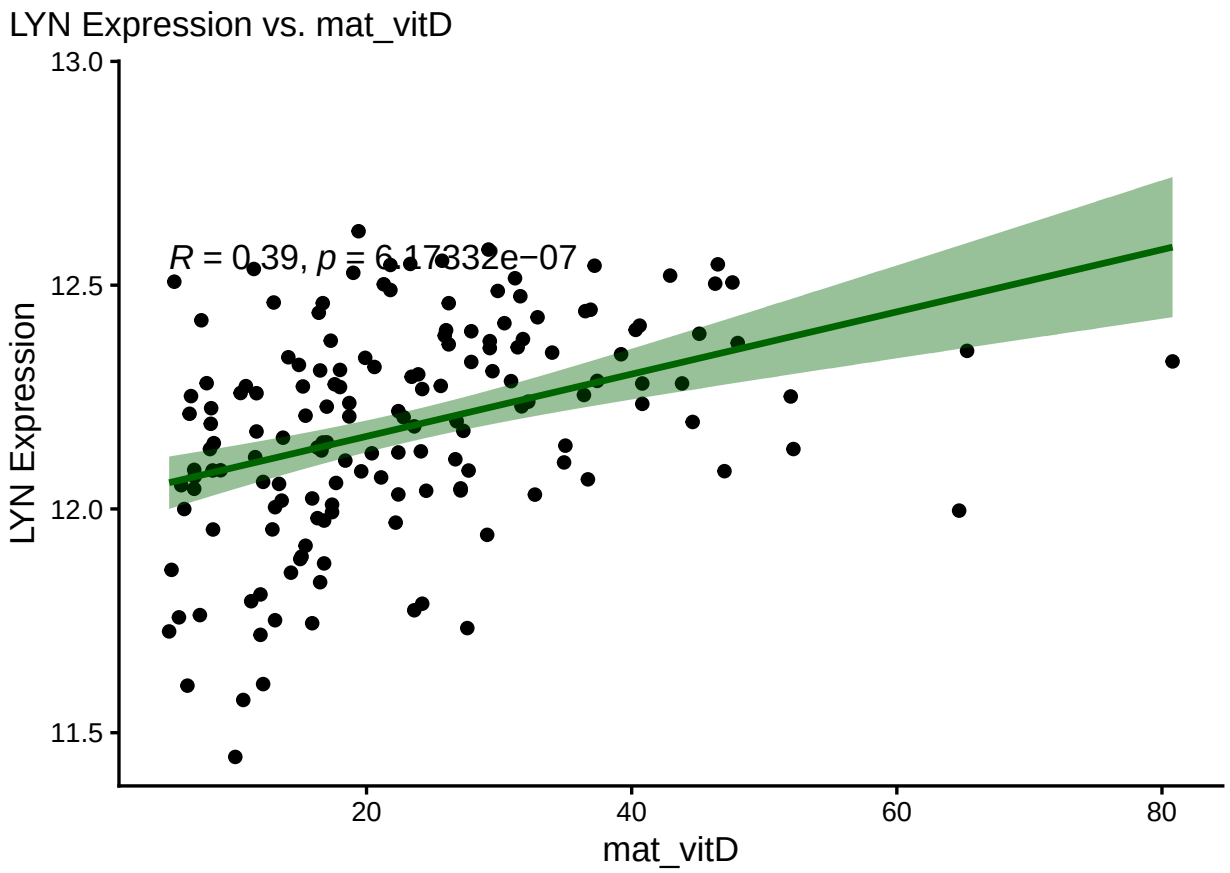


Plots for LYN

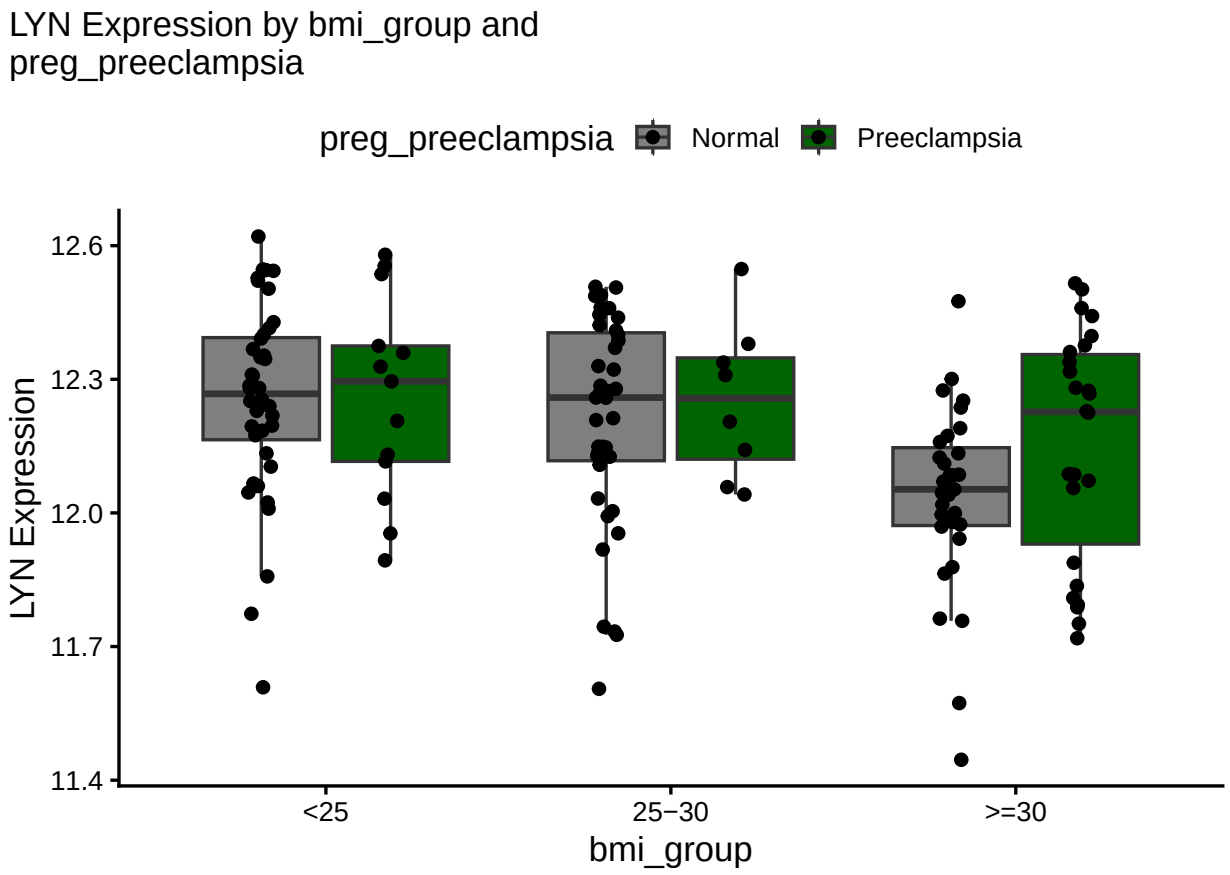
```
genePlots$LYN$histogram
```




```
genePlots$LYN$scatter
```

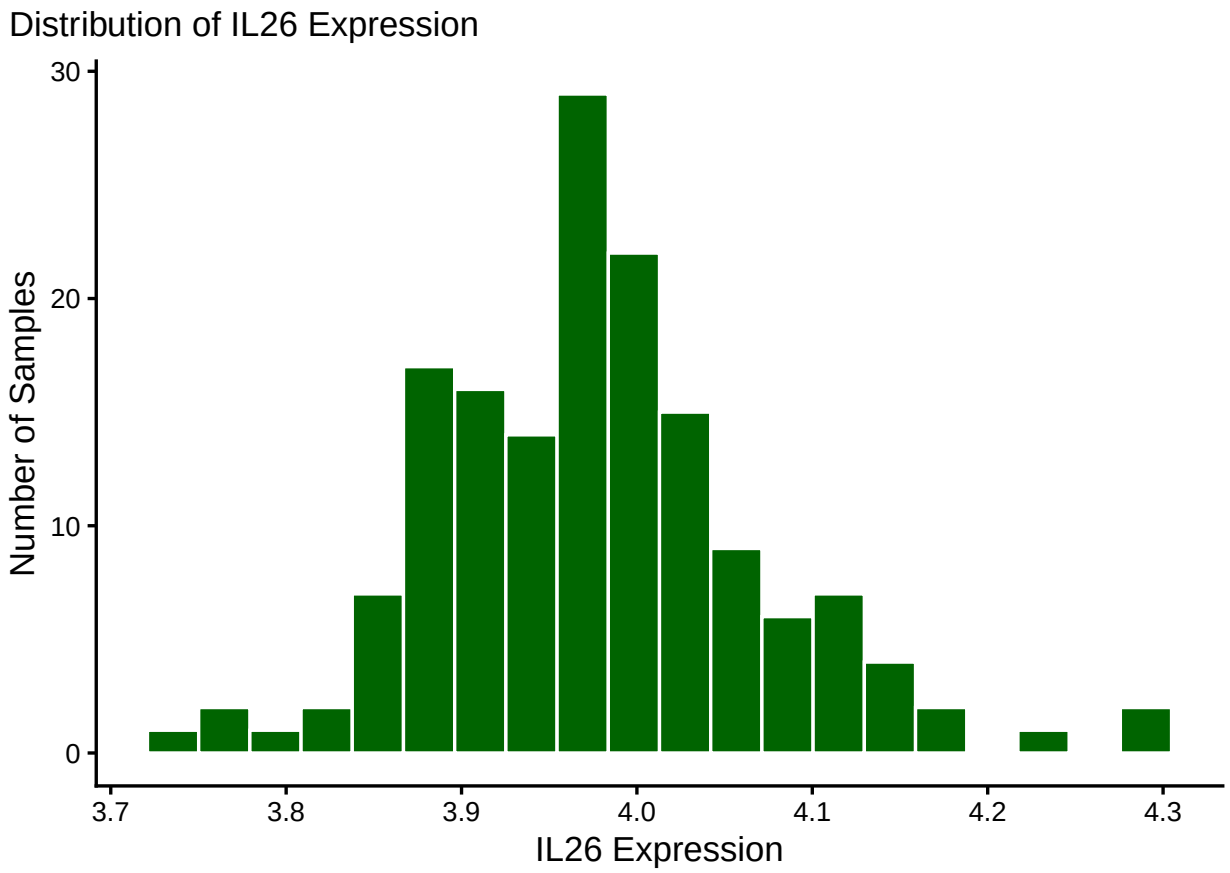



```
genePlots$LYN$boxplot
```

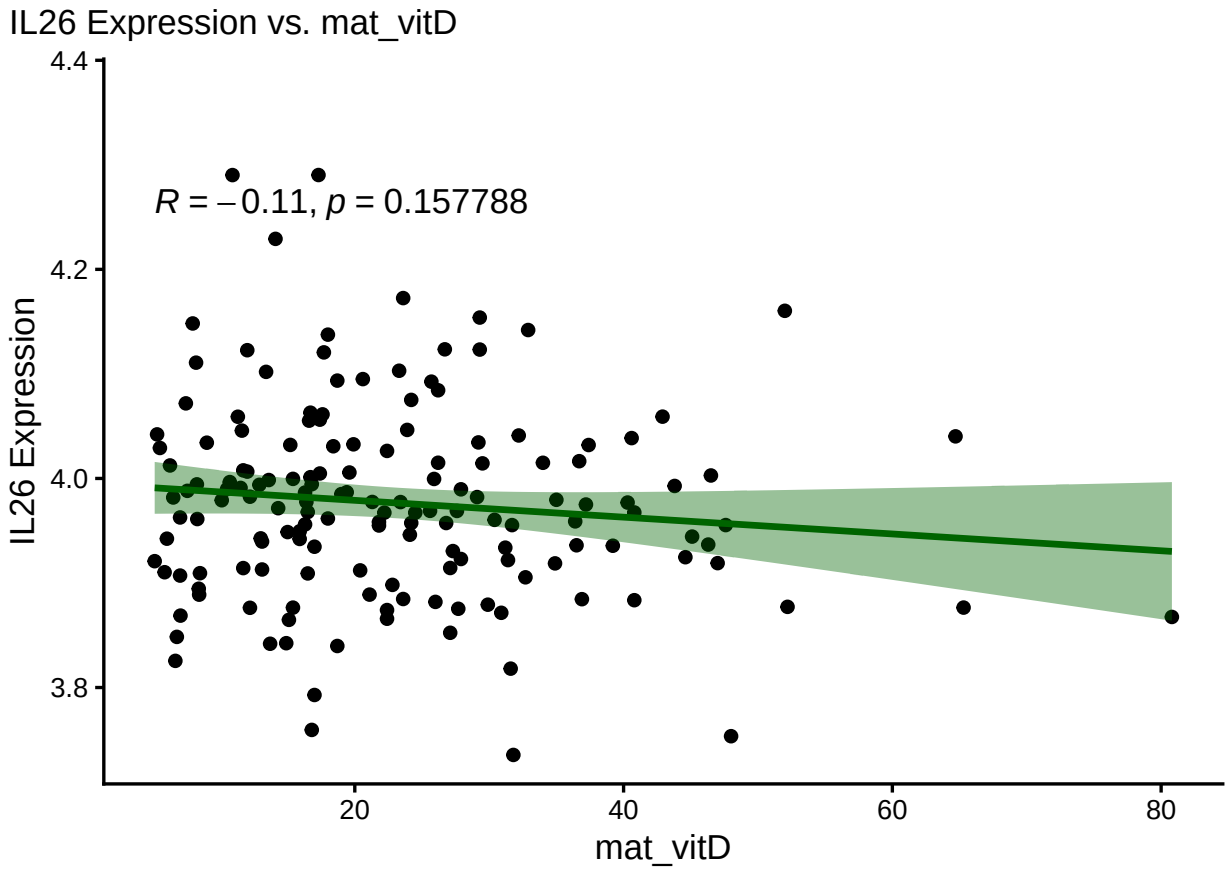


Plots for IL26

```
genePlots$IL26$histogram
```




```
genePlots$IL26$scatter
```




```
genePlots$IL26$boxplot
```

IL26 Expression by bmi_group and preg_preeclampsia

